

FlowAccel

Server Specification & Deployment Architecture

Prepared For:

Client Infrastructure Team

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1.1

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Confidential

1. Executive Summary

FlowAccel is an enterprise workflow management dashboard that integrates with the JotForm Workflow Engine API to manage multi-level approval workflows across 36 forms and 6,800+ submissions. It currently runs on Vercel serverless infrastructure. This document defines the on-premise server specification required to deploy FlowAccel on Windows IIS infrastructure, replacing the Vercel serverless architecture with a self-hosted Node.js backend.

Item	Value
Application	FlowAccel Workflow Dashboard
Frontend	React 18 + TypeScript + Vite (2.4 MB static build)
Backend	Node.js Express (converted from 22 Vercel serverless functions)
Database	PostgreSQL 15+ (self-hosted on-premise on Server 1)
External APIs	JotForm Enterprise API (eforms.mediaoffice.ae), Microsoft Azure AD
Servers Required	2 × Windows Server (Production + Backup)
Total Storage	500 GB per server (1 TB total)

2. Server Hardware Specification

2.1 Physical / Virtual Machine Specification

Component	Minimum	Recommended
CPU	4 vCPU (Intel Xeon E5 2.4 GHz or equivalent)	8 vCPU (Intel Xeon Gold 3.0 GHz+)
RAM	16 GB DDR4 ECC	32 GB DDR4 ECC
OS Disk	100 GB SSD (SATA) — Windows Server OS + IIS + Node.js	200 GB NVMe SSD — OS + application + logs
Data Disk	500 GB SSD — application data, uploads, backups, logs	500 GB NVMe SSD — high I/O for logs and uploads
Network	1 Gbps NIC	10 Gbps NIC (especially if serving external users)
Redundancy	RAID 1 for OS disk	RAID 1 for OS, RAID 5 for data
Power	Dual PSU (if physical)	Dual PSU + UPS

Important: Hyper-threading should be enabled. If virtualized (VMware/Hyper-V), ensure CPU and RAM reservation, not shared allocation. The Node.js event loop is single-threaded but benefits from multiple cores for concurrent requests and worker threads.

2.2 Server Count & Roles

Server	Role	Hostname (example)	Purpose
Server 1	Production	FLOWACCEL-PROD-01	Primary application server — IIS + Node.js backend + static content
Server 2	Backup / Staging	FLOWACCEL-BACKUP-01	Backup server — replicated configuration for disaster recovery; also serves as pre-production staging environment

High Availability: If zero-downtime failover is required, add a load balancer (hardware or Windows NLB) in front of both servers in an active-passive or active-active configuration. This specification covers the two-server baseline. For active-active, double the RAM to 32 GB per server.

3. Software Stack Requirements

3.1 Operating System

Component	Version	Notes
Windows Server	2022 Standard Edition (64-bit)	Or Windows Server 2019 with latest updates. Datacenter edition if running >2 VMs per host.
Windows Updates	Fully patched	All critical and security updates applied before deployment.

3.2 Web Server — IIS

Component	Required	Purpose
IIS 10	Yes (built-in with Windows Server 2022)	Web server — hosts static React files and reverse-proxies API requests
IIS URL Rewrite Module 2.1	Required	SPA routing — rewrites all non-file paths to index.html; also used for reverse proxy rules
IIS Application Request Routing (ARR) 3.0	Required	Reverse proxy module — forwards /api/* requests to the local Node.js process
IISNode	Required	Hosts Node.js application within IIS process, providing process management, auto-restart, and Windows-integrated logging
WebSocket Protocol	Enable	Required for real-time features and live data subscriptions

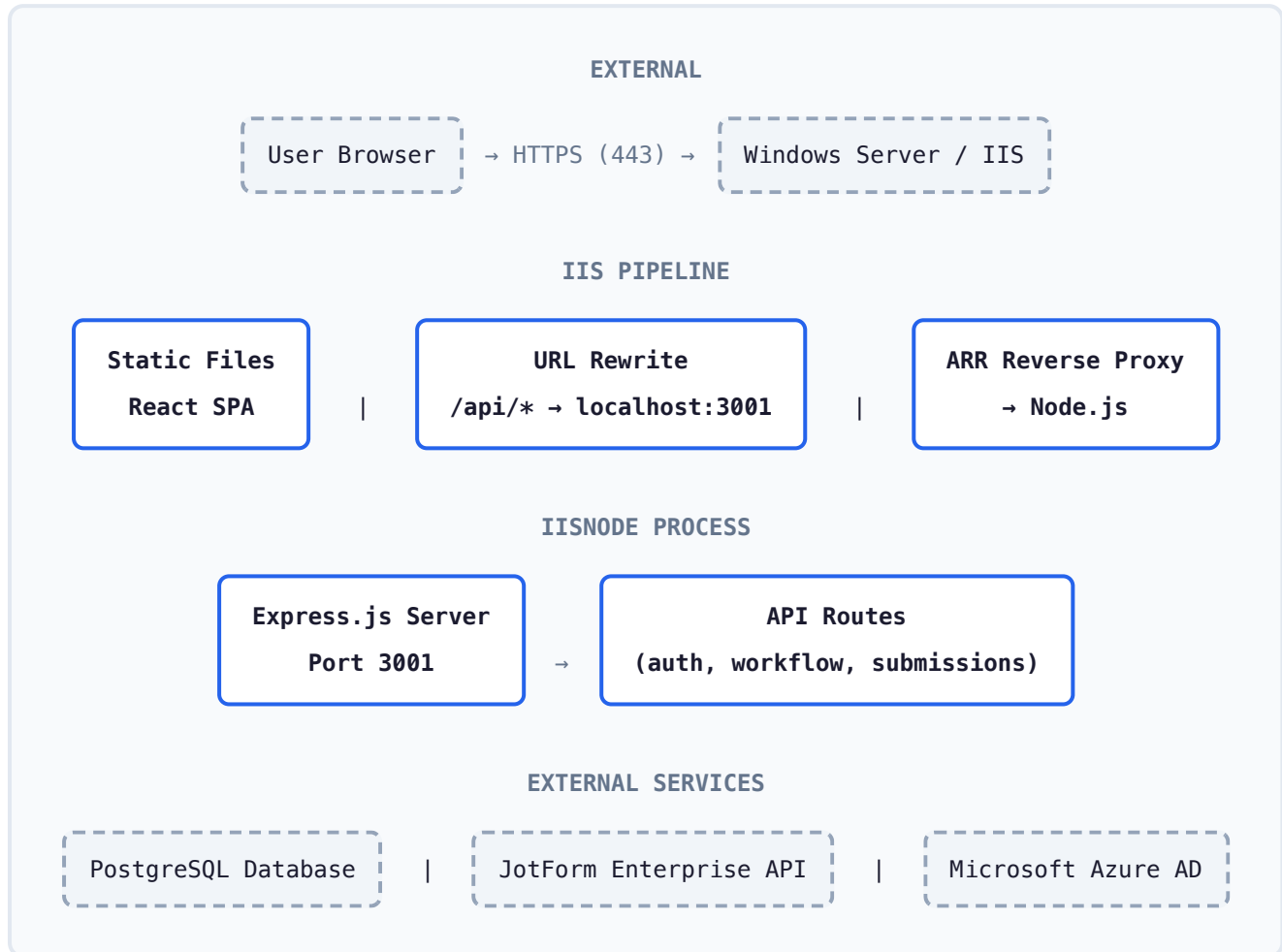
3.3 Runtime Dependencies

Component	Version	Purpose
Node.js LTS	18.x or 20.x LTS	JavaScript runtime for the backend API server
npm	9.x+	Package manager (bundled with Node.js)
URL Rewrite Module	2.1	Download from Microsoft IIS.net
ARR 3.0	3.0	Download from Microsoft IIS.net
IISNode	0.2.21+	Download from GitHub: azure/iisnode
Visual C++ Redistributable	2015-2022	Required by Node.js native modules; install both x86 and x64

Installation Order: (1) Windows Updates → (2) IIS Role + WebSocket → (3) URL Rewrite → (4) ARR 3.0 → (5) Visual C++ Redist → (6) Node.js → (7) IISNode → (8) Deploy Application.

4. Application Architecture on IIS

4.1 Architecture Diagram



4.2 Request Flow

- Browser** sends HTTPS request to `https://flowaccel.work`
- IIS** receives request on port 443 (HTTPS with SSL certificate)
- Static file:** If the request matches a file on disk (.js, .css, .png, etc.), IIS serves it directly from the `dist/` folder
- SPA route:** If the path does not match a file (e.g., `/dashboard`), URL Rewrite sends `index.html` — the React SPA handles client-side routing
- API request:** If the path starts with `/api/*`, ARR reverse-proxies the request to `http://localhost:3001/api/*` where the Express.js server handles it
- Express server** calls external services (JotForm API, Azure AD) and queries the local PostgreSQL database, returning JSON responses back through IIS to the browser

4.3 Key Architectural Change: Vercel Serverless → Express.js

The current application uses 22 Vercel serverless functions (TypeScript files in `/api`). For IIS deployment, these must be consolidated into a single Express.js application (~3,076 lines of API logic across 22 routes). The code conversion is straightforward — each serverless handler becomes an Express route handler with identical business logic:

```
// Current: Vercel serverless (api/workflow-action.ts)
export default async function handler(req: VercelRequest, res: VercelResponse) { ... }

// New: Express.js route (server.js)
app.post('/api/workflow-action', async (req, res) => { ... })
```

5. IIS Configuration Details

5.1 IIS Site Configuration

Setting	Value
Site Name	FlowAccel
Physical Path	C:\inetpub\wwwroot\flowaccel\dist
Application Pool	FlowAccelAppPool (.NET CLR Version: No Managed Code)
Bindings	HTTPS : 443 : flowaccel.work (with SSL certificate)
Authentication	Anonymous Authentication: Enabled; All others: Disabled
Default Document	index.html

5.2 Application Pool Configuration

Setting	Value	Reason
.NET CLR Version	No Managed Code	Node.js application, not .NET
Managed Pipeline Mode	Integrated	Required for URL Rewrite and ARR
Identity	NetworkService (or custom domain account)	Needs outbound internet access for JotForm API and Azure AD; needs local access to PostgreSQL
Idle Time-out	0 (Never time out)	Prevents Node.js process from being terminated during idle periods
Recycling — Regular Time Interval	0 (Disabled)	Let IISNode manage process lifecycle
Recycling — Private Memory Limit	1,048,576 KB (1 GB)	Recycle if Node.js exceeds 1 GB memory
Rapid-Fail Protection	Disabled	Prevents IIS from disabling the app pool if Node.js restarts during deployment
Start Mode	AlwaysRunning	Ensures Node.js starts immediately on server boot

5.3 IIS Node.js Process Management (IISNode)

Create `iisnode.yml` in the application root:

```
# iisnode.yml - Node.js process configuration
nodeProcessCountPerApplication: 0           # One process per CPU core (0 = auto)
maxConcurrentRequestsPerProcess: 1024       # Max concurrent HTTP requests per worker
node_env: production
loggingEnabled: true
logDirectory: C:\inetpub\logs\flowaccel\iisnode
maxLogFileSizeInKB: 10240                   # 10 MB per log file
maxTotalLogFileSizeInKB: 524288             # 500 MB total logs
devErrorsEnabled: false                     # Never show stack traces in production
flushResponse: true
watchedFiles: web.config;*.js;iisnode.yml
gracefulShutdownTimeout: 10000              # 10s grace period for in-flight requests
```

5.4 web.config — Complete IIS Configuration

```
<?xml version="1.0" encoding="UTF-8"?>
<configuration>
  <system.webServer>

    <!-- IISNode: Route server.js through IISNode module -->
    <handlers>
      <add name="iisnode" path="server.js" verb="*" modules="iisnode"
        resourceType="Unspecified" requireAccess="Script" />
    </handlers>

    <!-- Default document -->
    <defaultDocument>
      <files>
        <clear />
        <add value="index.html" />
      </files>
    </defaultDocument>

    <!-- URL Rewrite Rules -->
    <rewrite>
      <rules>

        <!-- Rule 1: API requests → reverse proxy to Node.js -->
        <rule name="API Proxy" stopProcessing="true">
          <match url="^api/(.*)" />
          <action type="Rewrite" url="http://localhost:3001/api/{R:1}" />
        </rule>

        <!-- Rule 2: Static files → serve directly if exists -->
        <rule name="Static Content" stopProcessing="true">
          <match url="(.*)" />
          <conditions logicalGrouping="MatchAny">
            <add input="{REQUEST_FILENAME}" matchType="IsFile" />
          </conditions>
        </rule>

      </rules>
    </rewrite>
  </system.webServer>
</configuration>
```

```
        <add input="{REQUEST_FILENAME}" matchType="IsDirectory" />
    </conditions>
    <action type="None" />
</rule>

<!-- Rule 3: SPA fallback – all other routes → index.html -->
<rule name="SPA Fallback" stopProcessing="true">
    <match url="(.*)" />
    <action type="Rewrite" url="/index.html" />
</rule>

</rules>
</rewrite>

<!-- Security headers -->
<httpProtocol>
    <customHeaders>
        <add name="X-Content-Type-Options" value="nosniff" />
        <add name="X-Frame-Options" value="DENY" />
        <add name="X-XSS-Protection" value="1; mode=block" />
        <add name="Referrer-Policy" value="strict-origin-when-cross-origin" />
        <add name="Strict-Transport-Security" value="max-age=31536000; includeSubDomains" />
    </customHeaders>
</httpProtocol>

<!-- CORS: allow frontend origin -->
<httpProtocol>
    <customHeaders>
        <add name="Access-Control-Allow-Origin" value="https://flowaccel.work" />
    </customHeaders>
</httpProtocol>

<!-- Gzip compression for static assets -->
<urlCompression doStaticCompression="true" doDynamicCompression="true" />

<!-- Cache static assets for 30 days -->
<staticContent>
    <clientCache cacheControlMode="UseMaxAge" cacheControlMaxAge="30.00:00:00" />
</staticContent>

</system.webServer>
</configuration>
```

6. Application Deployment Steps

6.1 Directory Structure on Server

```
C:\inetpub\wwwroot\flowaccel\  
├─ dist/                                # Vite build output (static React SPA)  
│   ├── index.html  
│   └─ assets/  
│       ├── index-*.js  
│       ├── index-*.css  
│       └─ ...  
├─ server.js                            # Express.js backend (converted from /api)  
├─ node_modules/                        # Production dependencies only  
├─ iisnode.yml                          # IISNode configuration  
├─ web.config                           # IIS configuration  
├─ .env                                 # Environment variables (secured)  
└─ logs/                                # Application logs
```

6.2 Environment Variables (.env)

Variable	Description	Example Value
JOTFORM_API_KEY	JotForm Enterprise API key	<i>(secure — provided separately)</i>
JOTFORM_TEAM_ID	JotForm team ID (leave empty for full access)	
DATABASE_URL	PostgreSQL connection string	postgresql://user:password@localhost:5432/flowaccel
DB_HOST	PostgreSQL host	localhost
DB_PORT	PostgreSQL port	5432
DB_NAME	PostgreSQL database name	flowaccel
DB_USER	PostgreSQL user	flowaccel_app
DB_PASSWORD	PostgreSQL password	<i>(secure — provided separately)</i>
AZURE_CLIENT_ID	Microsoft Azure AD app registration client ID	<i>(secure — provided separately)</i>
AZURE_TENANT_ID	Microsoft Azure AD tenant ID	<i>(secure — provided separately)</i>
AZURE_CLIENT_SECRET	Microsoft Azure AD client secret	<i>(secure — provided separately)</i>
ALLOWED_ORIGIN	CORS origin	https://flowaccel.work
PORT	Node.js backend port (internal only)	3001
NODE_ENV	Runtime environment	production

Security: The `.env` file must be protected with strict NTFS permissions — only the Application Pool identity and Administrators should have read access. Never commit this file to source control or expose it via IIS.

6.3 Build & Deployment Commands

```
# 1. Build the React frontend
npm install
npm run build                # Runs tsc -b && vite build → outputs to dist/

# 2. Install production dependencies only
npm ci --production          # ~15–20 MB for production deps

# 3. Copy to IIS directory
robocopy dist C:\inetpub\wwwroot\flowaccel\dist /E /MIR
copy server.js C:\inetpub\wwwroot\flowaccel\
copy web.config C:\inetpub\wwwroot\flowaccel\
copy iisnode.yml C:\inetpub\wwwroot\flowaccel\
copy .env C:\inetpub\wwwroot\flowaccel\

# 4. Set NTFS permissions (PowerShell)
icacls C:\inetpub\wwwroot\flowaccel\.env /inheritance:r /grant "IIS AppPool\FlowAccelAppP
icacls C:\inetpub\wwwroot\flowaccel\logs /grant "IIS AppPool\FlowAccelAppPool":(M)

# 5. Restart IIS site
iisreset /restart
```

7. SSL Certificate & HTTPS Configuration

7.1 SSL Certificate Requirements

Requirement	Details
Certificate Type	Standard SSL (DV) or Organization Validation (OV) SSL
Domain(s)	flowaccel.work (primary); optionally www.flowaccel.work as SAN
CA Recommendations	DigiCert, Sectigo, or Let's Encrypt (automated renewal via win-acme)
Key Length	RSA 2048-bit minimum; ECDSA P-256 preferred
TLS Version	TLS 1.2 minimum; TLS 1.3 preferred (disable TLS 1.0/1.1)
Renewal	90-day auto-renewal with win-acme (Let's Encrypt) or manual annual renewal

7.2 IIS SSL Binding (PowerShell)

```
# Import certificate (PFX format)
Import-PfxCertificate -FilePath C:\certs\flowaccel.pfx `
  -CertStoreLocation Cert:\LocalMachine\My -Password (Read-Host -AsSecureString)

# Add HTTPS binding
New-IISSiteBinding -Name "FlowAccel" `
  -BindingInformation "*:443:flowaccel.work" `
  -CertificateThumbprint "THUMBPRINT_HERE" `
  -CertStoreLocation "Cert:\LocalMachine\My" `
  -Protocol https
```

Auto-renewal with Let's Encrypt: Install [win-acme](#) (Windows ACME client). Configure it to auto-renew every 60 days and update the IIS binding automatically. This eliminates manual certificate renewal.

8. Database Architecture

8.1 PostgreSQL — Self-Hosted on Server 1

FlowAccel uses **PostgreSQL 15+** installed directly on the production server. The database stores workflow submissions, approval history, approver configurations, and notifications. The Express.js

backend connects to PostgreSQL via the `pg` Node.js driver using a local connection string.

Component	Details
Database	PostgreSQL 15+ (self-hosted on Server 1)
Connection	Local TCP connection (localhost:5432) via pg driver
Tables	jf_submissions, jf_approval_history, jf_approver_config, notifications, user_profiles
Auth	JWT-based authentication with bcrypt password hashing; Microsoft Azure AD SSO via OAuth 2.0
Data Volume	~6,800+ submissions, ~1,500+ approval records, growing
Port	5432 (local only — not exposed to network)

8.2 PostgreSQL Installation on Windows Server

```
# 1. Download PostgreSQL 15 installer from https://www.postgresql.org/download/windows/
# 2. Run installer with following options:
#   - Install directory: C:\Program Files\PostgreSQL\15
#   - Data directory: D:\PostgreSQL\15\data (on the 500 GB data disk)
#   - Port: 5432
#   - Locale: English, UTF-8

# 3. Create application database and user (PowerShell as Administrator)
cd "C:\Program Files\PostgreSQL\15\bin"
psql -U postgres -c "CREATE DATABASE flowaccel;"
psql -U postgres -c "CREATE USER flowaccel_app WITH ENCRYPTED PASSWORD 'strong_password_h"
psql -U postgres -c "GRANT ALL PRIVILEGES ON DATABASE flowaccel TO flowaccel_app;"

# 4. Configure pg_hba.conf – allow local connections only
# Add to C:\Program Files\PostgreSQL\15\data\pg_hba.conf:
# host    flowaccel    flowaccel_app    127.0.0.1/32    scram-sha-256

# 5. Run schema setup script (provided in deployment package)
psql -U flowaccel_app -d flowaccel -f schema.sql

# 6. Configure PostgreSQL service for automatic startup
sc config postgresql-x64-15 start= auto
```

8.3 Table Schema Summary

Table	Purpose	Key Columns
jf_submissions	Core submission data from JotForm	jotform_submission_id (PK), form_id, title, status, current_level, pending_approver_email, submitter_email, submission_date
jf_approval_history	Audit trail for workflow approvals	id (PK), submission_id (FK), level, action, approver_name, approver_email, comment, created_at
jf_approver_config	Manual approver assignments per form/level	id (PK), form_id, level, approver_name, approver_email
notifications	User notification inbox	id (PK), user_email, type, title, message, read, created_at
user_profiles	User profiles and auth data	id (PK), email, name, role, azure_oid, password_hash, created_at

PostgreSQL Configuration: Set `shared_buffers = 512MB` , `effective_cache_size = 4GB` , and `work_mem = 16MB` in `postgresql.conf` for optimal performance with the expected workload. Enable `pg_stat_statements` extension for query performance monitoring.

9. Firewall & Network Requirements

9.1 Inbound Rules (Internet → Server)

Port	Protocol	Source	Destination	Purpose
443	TCP	Any (or specific IP ranges if VPN)	Server IP	HTTPS — user access to the application
3389	TCP	Admin IP range only	Server IP	RDP — remote administration (restrict tightly)

9.2 Outbound Rules (Server → Internet)

Port	Protocol	Destination	Purpose
443	TCP	eforms.mediaoffice.ae	JotForm Enterprise API — fetch forms, submissions, workflow data
5432	TCP	localhost only (127.0.0.1)	PostgreSQL — local database connection (not exposed to network)
443	TCP	login.microsoftonline.com	Microsoft Azure AD OAuth token exchange
443	TCP	graph.microsoft.com	Microsoft Graph API — user profile lookup (optional)
80 / 443	TCP	acme-v02.api.letsencrypt.org	SSL certificate auto-renewal (if using Let's Encrypt)

Critical: If the client's network uses an HTTP proxy for outbound traffic, the Node.js process must be configured to use it via `HTTP_PROXY` and `HTTPS_PROXY` environment variables. All outbound API calls are HTTPS-only.

10. Monitoring & Maintenance

10.1 Log Files

Log Type	Location	Retention
IIS Access Logs	C:\inetpub\logs\LogFiles\W3SVC*	Rolling 90 days (compress after 30)
IISNode Logs	C:\inetpub\logs\flowaccel\iisnode\	Rolling 30 days
Application Logs	C:\inetpub\wwwroot\flowaccel\logs\	Rolling 60 days
Windows Event Log	Event Viewer → Application	System default
Failed Request Traces	C:\inetpub\logs\FailedReqLogFiles\	Enable on-demand for debugging

10.2 Health Checks

Endpoint	Expected	Frequency
GET /api/jotform?path=user/forms	HTTP 200 with JSON response	Every 5 minutes
GET /	HTTP 200, returns index.html	Every 1 minute
Node.js process running	1+ iisnode.exe processes	Continuous (IISNode auto-monitors)

10.3 IIS App Pool Monitoring Script (PowerShell)

```
# check-flowaccel.ps1 – Health check script (run as scheduled task every 5 min)
$response = Invoke-WebRequest -Uri http://localhost:3001/api/jotform?path=user/forms -Use
if ($response.StatusCode -ne 200) {
    Write-EventLog -LogName Application -Source "FlowAccel" -EventId 1001 `
        -EntryType Error -Message "FlowAccel API health check failed"
}
```

11. Backup Strategy

What	Method	Frequency	Retention
Application Code & Config	Robocopy to Backup server + offsite	After each deployment	Keep last 5 versions
IIS Configuration	<code>appcmd list config - section:system.webServer</code> output	Weekly	3 months
SSL Certificate	Export PFX with private key	On renewal	Keep current + previous
.env Configuration	Encrypted copy to secure vault	On any change	Permanent
Windows Server	Windows Server Backup (wbadmin) to network share	Daily incremental, weekly full	4 weekly full backups

PostgreSQL backups: Schedule daily backups via Windows Task Scheduler using `pg_dump`. Store backup files on the 500 GB data disk and replicate to the Backup server weekly. Command:

```
"C:\Program Files\PostgreSQL\15\bin\pg_dump.exe" -U flowaccel_app -d flowaccel -Fc -f "D:\Backups\flowaccel_%date:~4,4%%date:~10,2%%date:~7,2%.dump"
```

Retention: 7 daily + 4 weekly backups.

12. Estimated Resource Utilization & Costs

12.1 Expected Server Load

Metric	Value
Concurrent Users	50–200 (approvers + admins within the organization)
Requests per Minute	~100–500 (dashboard loads, API calls, real-time sync)
Average Response Time	< 200ms for API calls; < 2s for full dashboard load
Memory Usage (Node.js)	150–350 MB per IISNode worker process (2–4 workers)
Total RAM Usage	~2 GB (IIS + Node.js) + OS overhead (~3 GB) = ~5 GB baseline
Disk Growth Rate	~500 MB/month (logs + uploaded signatures)
Outbound Bandwidth	~50–200 GB/month (JotForm API data, submission enrichment)

12.2 Estimated Licensing Costs (Indicative)

Item	Qty	Est. Annual Cost (USD)
Windows Server 2022 Standard (16-core)	2	\$1,948 (\$974 × 2)
SSL Certificate (DV)	1	\$0 (Let's Encrypt) or ~\$200/year (Sectigo/DigiCert)
PostgreSQL	1	\$0 (open-source, self-hosted on Server 1)
JotForm Enterprise	1	Already licensed (existing GDMO contract)
Total Annual (Excluding Hardware)		~\$1,948–\$2,148

Note: Hardware costs are not included as they depend on whether the servers are physical, virtualized on existing infrastructure, or cloud VMs (Azure/AWS). If using Azure VMs (D4s v3, 4 vCPU, 16 GB RAM), the estimated cost is ~\$310/month per VM (~\$7,440/year for 2 VMs).

13. Support & Handover Requirements

Deliverable	Provided By	Timeline
Express.js backend server.js (converted from Vercel functions)	Development Team	2-3 business days after server provisioning
web.config + iisnode.yml configuration files	Development Team	Included in deployment package
Environment variable values (secure handover)	Development Team	Via encrypted channel on deployment day
Windows Server + IIS installation	Client IT Team	Before deployment date
SSL certificate procurement and installation	Client IT Team	Before deployment date
Firewall rule configuration	Client IT Team	Before deployment date
NTFS permissions + security hardening	Client IT Team	After deployment

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